CAS Harness AI2 One-Pager

Moniker: Turn your black box into a glass house.

One-sentence claim: CAS Harness is an open-weight AI audit system that reads model-internal CAS structure, tests whether that structure is causally load-bearing, and returns an ACCEPT / SUPPRESS / FLAG verdict with an Evidence Packet.

What CAS Harness Does

1. Connect an open-weight model.
2. Capture hidden states across layers.
3. Probe for CAS taxonomies that were defined before the model run.
4. Test whether those signals survive controls and collapse under targeted removal.
5. Measure whether CAS-conditioned interventions move model output in the predicted direction.
6. Return a verdict with confidence, proof chips, and a trace-id-keyed Evidence Packet.

The product is not just a visualization layer. Atlas Viewer is the cockpit. CAS Harness is the audit engine and evidence protocol underneath it.

Core Mechanism: Eta To Epsilon Bridge

Eta: read CAS structure from the foreign model's hidden states.

Epsilon: realize that detected CAS structure as readable English prefix tokens.

Bridge: run the model once, read the CAS state, prepend the readable CAS realization, then run the model again and measure whether the likelihood shifts in the CAS-predicted direction.

This matters because ablation-based generation steering repeatedly failed. Prefix realization worked. The reason is architectural: later transformer positions can attend to prefix positions by construction. CAS uses that property instead of pretending that every hidden-state ablation should directly steer generation.

What Is Already Closed

Evidence tier	Result	Why it matters
28 Foundations readout	Pythia-160M: 78.6%, +28.9 points over text baseline	CAS reads a model-internal Foundation structure above surface-text guessing.

Evidence tier	Result	Why it matters
40 Elements readout	Pythia-160M: 89.8%, +39.7 points over text baseline	The strongest original readout used the TKS 40 Element grid.
Causal subspace ablation	28 Foundations: 78.6% -> 3.57% exact 1-in-28 chance	Removing the CAS subspace kills the signal.
Random controls	Same-rank random ablations: mean disruption 0.000, max 0.000	The collapse is not explained by generic damage.
Likelihood bridge	Pythia-160M +0.283, Pythia-1B +0.554, OLMo-1B +0.604, Qwen2.5-1.5B +0.796	CAS conditioning changes output likelihood in the predicted direction across model families.
Raw-symbol control	Qwen raw-CAS-symbol control: -0.0002	The effect requires readable epsilon realization, not decorative CAS symbols.
Verification Stack	OLMo: 98.16% accepted accuracy at 71.73% coverage; Qwen: 99.49% at 74.57% coverage	CAS separates high-confidence accepted rows from suppress / review rows.
Weight geometry	Pythia-2.8B: 2.83x random alignment, p = 0.0099	Static output-weight evidence independently agrees with the dynamic probe story.

Fresh 20B Status

Model: GPT-NeoX-20B, real open weights, 44 transformer layers.

What passed:

Run	Dataset	Readout	Controls	Targeted ablation
TKS 40 Elements	400 rows, 40 classes	40.00%	2.50% chance / 2.56% shuffled	40.00% -> 26.25%; random/wrong controls approximately zero
Exact original Stage 7 Foundation-28	4,368 rows, 156 per class	51.04%	3.57% chance / 3.40% shuffled / 42.28% text	51.04% -> 38.59%; random/wrong controls approximately zero

Plain English: the 20B model is not just echoing labels. It carries measurable TKS/CAS structure internally, and targeted cover removal weakens that structure while random and wrong-cover removals do not.

Fresh behavioral bridge result: the exact Stage 7 eta-to-epsilon likelihood bridge was run on GPT-NeoX-20B with 112 same-register drive-cycle cases. Under the original balanced-logistic readout methodology, the probe-predicted readable epsilon loop passes: `+0.596413` mean likelihood-margin delta, `90/112` positive rows, sign-test p = `3e-11`, and `+1.244917` separation from the wrong readable partner. Gold/readable epsilon prefixes remain strong: `+0.660065`, `93/112` positive rows. Raw CAS-symbol prefixes stay near zero (`+0.002524` probe raw, `+0.002186` gold raw), confirming the load-bearing route is readable epsilon realization, not decorative symbols.

Methodology correction: the earlier weak bridge used nearest-centroid readout, which was not the original closure probe family. Replacing it with sklearn `LogisticRegression(C=0.1, class_weight="balanced")` after `StandardScaler` raises heldout Foundation readout to `86.8622%`, selected bridge-row Foundation accuracy to `85.7143%`, and selected Drive accuracy to `93.75%`.

Plain English: the 20B bridge works when we use the original probe method. The failure was not the TKS definitions and not the eta-to-epsilon mechanism. It was a readout implementation mismatch: centroid was too weak; balanced logistic is the correct Stage 7 readout.

Fresh trace diagnostic result: the exact Stage 7 Foundation-28 trace also passes the lacunarity/fractal preflight on GPT-NeoX-20B. At bridge layer 40, heldout Foundation accuracy is `86.8622%`, Foundation lacunarity is `0.006715` against a pass threshold of `<= 0.05`, drive lacunarity is `0.002752`, register lacunarity is `0.001283`, and the layer-trace preflight class is `Amplifying`. For

comparison, the original Pythia-160M Foundation lacunarity was `0.021`, so the 20B Foundation trace is cleaner on this gate.

What is not closed yet on 20B: this is now a probe-predicted behavioral likelihood-bridge pass plus a trace-diagnostic pass, but not yet a full CAS Harness ACCEPT/SUPPRESS verdict. The next gate is a fresh Evidence Packet verdict and final verifier merge. The fractal result above is a numeric layer-trace proxy, not yet the discrete `cas-engine` orbit classifier.

Why This Is Different From Existing Interpretability Tools

Neuronpedia, TransformerLens, nnsight, and attribution-graph tools are useful. CAS Harness is aiming at a different product line:

- It turns interpretation into an audit verdict: ACCEPT, SUPPRESS, or FLAG.
- It binds every verdict to an Evidence Packet instead of leaving it as a notebook result.
- It tests readout, causal removal, matched controls, output prediction, and replication as one protocol.
- It uses source-defined TKS/CAS taxonomies instead of post-hoc feature naming.
- It is being extended from small models to 20B+ open-weight models with the same methodology.

What Would Prove Universality

CAS universality is not proven by a pretty cockpit or by one successful probe. The standard is:

1. Same source-derived TKS/CAS taxonomies.
2. Same hidden-state readout protocol.
3. Same matched controls.
4. Same targeted ablation / cover-removal test.
5. Same eta-to-epsilon likelihood bridge.
6. Same Evidence Packet format.
7. Replication across model families and sizes, including 20B, 100B, and 300B open-weight systems.

What We Want AI2 To React To

1. Is the corrected TKS 28 Foundation / 40 Element methodology scientifically credible?
2. What would AI2 consider the cleanest 20B closure threshold?
3. What model family should be the next serious replication target after GPT-NeoX-20B?
4. What infrastructure would make 100B / 300B open-weight CAS Harness audits practical?

5. What Evidence Packet fields would senior researchers need before treating a CAS verdict as audit-grade?

Honest Boundary

CAS Harness has strong closed evidence on Pythia, OLMo, and Qwen, plus fresh 20B evidence: exact Stage 7 Foundation readout, targeted ablation, eta-to-epsilon behavioral bridge, and lacunarity/fractal trace preflight. It is not yet a completed 300B enterprise audit product. The next scientific step is to merge the 20B chain into a full Evidence Packet and verifier verdict.